

GRAD 50400 - Project Checkpoint-2

Team Project Design Review: Model Selection, Training, Evaluation

Due: April 19, 11:P9 PM Eastern Standard Time on **Brightspace**

1 To do

For this project checkpoint, you need to do these tasks:

1. Follow steps 3-4 and complete the model choosing, performance measure related tasks.
2. Prepare a writeup and submit it on Brightspace assignment portal
3. Prepare a presentation video and submit
 - (a) the video to the Brightspace discussion board for peer review
 - (b) the associated .pptx file to the Brightspace assignment portal
4. Prepare and submit the peer review form for the other group you are individually assigned to for reviewing in Discussion link (the link will be given in an announcement)

1.1 Writeup

As a team you will do a writeup on Steps 3 and 4. The writeup from Steps 1-4, including all proposals (and, if you desire, the teams preference and reasoning among competing proposals) will be submitted in the Brightspace. Please have a look at [3.2](#), [2.3](#) and [2.4](#) for clarity.

You need to submit a writeup file **must be** be a **.pdf**, exported from either

- From .ipynb file where the analyses and writings are done simultaneously, or
- From a .pptx file, where the analyses and writings are prepared as a slide, or
- From a .docx file, where the analyses and writings are prepared in a document, or
- From a .tex file, where the analyses and writings are prepared in L^AT_EX

1.2 Presentation

Your group should record a 15 minute presentation (recorded). Each student will be responsible for presenting their part.

Also, for presentation, you need to submit

1. The .pdf file exported from .pptx or LaTeX Beamer slides to Brightspace assignment portal, and
2. a .mov or .mp4 file with the recorded presentation to the discussion board (will be informed through a post)

1.3 Peer Review

You will individually be reviewing another group project. The review assignment is attached with the zip file (in case it is not there, it will be provided with an announcement). You will be provided with a rubric to assist in this evaluation, but you should feel to provide feedback you think the other teams will find helpful outside that rubric. The peer review is due on **April 21, 11:P9 PM Eastern Standard Time**. You need to submit the **.pdf** exported version of the Project review form. The form is attached with the zip file.

You can have a look at some of the samples provided from the previous courses.

You just need to submit one set of materials per group.

2 Overview-Project Steps

2.1 Step 1

Step 1 is to define a problem. While this is open, you should keep in mind that it should be a problem you reasonably could solve as a person (although perhaps not able to scale to the data sizes you'll use.) It must be possible to evaluate success - have you solved the problem? You also need to ensure that data exists to support learning a solution.

2.2 Step 2

Step 2 is data identification/exploration. You should identify at least two potential data sources. Establish that data is available: You can obtain it, and there are no intellectual property, privacy, or other ethical concerns that prevent you from doing this as a class project. You will then establish suitability of the data for the problem (data exploration, quality evaluation - week 2 of the course.)

2.3 Step 3 - Needed for this submission

Step 3 is a formal plan to evaluate model efficacy. You will have to identify appropriate metrics, explain why they are appropriate, decide what constitutes success, determine the data to be used for evaluation (does this come from the same source as the training data? Should it?), and a method for evaluation. Note that the metrics could involve many different things - simply accomplishing the task may not be enough - does it run fast enough for the problem at hand? Does it scale? You should have at least two, and possibly three, proposals for this. These could be competing proposals, or different aspects of the evaluation. Each proposal will be primarily the responsibility of a subtask lead, but should be a collaborative effort.

2.4 Step 4 - Needed for this submission

Step 4 is to identify specific modeling techniques. For any technique you propose, you should describe why it is appropriate, any tools you will use (e.g., from the python scikit-learn package, or if you will code up your own). Will you be making use of a foundation model or training from scratch? What are the computational needs? Other resources? As with Step 3, there should be at least two proposals lead by different team members.

2.5 Steps 5

Step 5 is to implement and evaluate at least two of your proposals. This will involve coding, building the models, and evaluating. You will then submit a writeup and 15 minute presentation discussing results: What worked, what didn't, and formal evaluation results. You should also discuss deployment - could you use this to solve the original problem? Pitfalls to watch for in deployment? Lessons learned? As with Steps 3 and 4, there should be at least two relatively independent, identifiable sub-tasks (building and evaluating the model, implementing and testing two competing proposals, etc.) The individuals responsible should be identified in the writeup. Integrate one of the feedbacks from peer review in your project and mention about it in the report and the presentation. Also, put the codebase in a public repository/drive and mention the link in both the report and the presentation.

3 CheckPoints

3.1 Check-In #1 - Team Project: Problem Definition, Data Exploration

You will submit a writeup on Steps 1 and 2, that will be evaluated to ensure the project looks feasible, and to provide early feedback to ensure it is. This writeup should give a statement of the problem, identify the data and discuss availability, and include an evaluation of the data and it's suitability for the problem. This writeup should also identify task leads for the remaining tasks.

3.2 Check-In #2 - Team Project Design Review: Model Selection, Training, Evaluation: [You are here](#)

You will do a writeup on Steps 3 and 4. The writeup from Steps 1-4, including all proposals (and, if you desire, the teams preference and reasoning among competing proposals) will be submitted to other students for review, along with a 15 minute presentation. Each student will be responsible for reviewing one project. You will be provided with a rubric to assist in this evaluation, but you should feel to provide feedback you think the other teams will find helpful outside that rubric.

3.3 Final - Team Project Final Write-up and Presentation

The writeup and presentation is due at the beginning of the final week of class. The final project submission process is the same as we used in check-in #2. One member of your team should submit the assignment for grading by the course instructor.

You should divide tasks so that each team member is responsible for two tasks from Steps 3-5.

Be sure to review the AI Use Policy.